

PSL Cricket Analysis

Predicting Final Score from the Powerplay

ML Assignment 1 | Linear Regression

Course: Data Analytics | Machine Learning

1. Introduction

In T20 cricket, the first 6 overs (known as the Powerplay) set the tone for the entire innings. During this phase, only two fielders are permitted outside the 30-yard circle, incentivising aggressive batting. This report presents a Linear Regression model trained to predict a PSL team's final innings total using only Powerplay statistics. The analysis uses real PSL delivery data spanning multiple seasons, aggregated to 717 innings-level observations.

2. Dataset Overview

Two datasets were provided and used in this analysis:

File	Rows	Description
psl_matches.csv	83,594	Ball-by-ball raw data. Each row is one delivery across all PSL seasons.
psl_innings_dataset.csv	717	Innings-level summary. One row per innings, used directly for modelling.

Key dataset findings:

- Total deliveries bowled across all PSL matches: 83,594
- Highest single-innings Powerplay total: 97 runs by Islamabad United
- Most Powerplay dot balls in a single innings: 28 by Peshawar Zalmi
- Average final innings total across all 717 innings: 160 runs
- Feature with the most statistical variation: final_total (std $\approx$ 29 runs)

3. Which Powerplay Feature Mattered Most?

The model was trained on seven features: powerplay_runs, powerplay_wickets, powerplay_run_rate, powerplay_sixes, powerplay_fours, powerplay_dot_balls, and innings. After fitting Linear Regression on 573 training innings, the coefficients revealed the following:

The combined effect of Powerplay run rate and Powerplay runs is by far the strongest predictor of the final total. **Note on multicollinearity:** Since $\text{powerplay_run_rate} = \text{powerplay_runs} \div 6$, these two features are nearly perfectly correlated ($r = 0.9999$). This causes the model to assign large opposite-sign coefficients (+739 and -711) that cancel each other out, a known statistical artifact called multicollinearity. Their combined direction is correct and cricket-sensible: the more runs scored in the Powerplay, the higher the predicted final total.

The remaining coefficients align with cricket logic:

- `powerplay_wickets`: negative coefficient (-10.5). Losing wickets in the Powerplay forces a rebuild and depresses the final score.
- `powerplay_dot_balls`: negative coefficient. Dot balls build pressure and reduce the balls available for big-hitting later.
- `innings`: negative coefficient (-10.5). Second-innings chasing sides typically finish with lower totals as they target the exact required runs rather than maximising scoring.

Coach takeaway: Focus the first 6 overs on boundary hitting and protecting top-order wickets. Every extra run scored in the Powerplay is strongly associated with a higher final total. Losing three or more wickets in the Powerplay is a significant handicap to the final score.

4. Model Accuracy

The model was evaluated on a held-out test set of 144 innings (20% of the dataset). The results are summarised below:

Metric	Value	What it means
MAE	22.7 runs	On average, predictions were off by 22.7 runs.
RMSE	29.2 runs	Root mean squared error; penalises large prediction errors more heavily.
R^2 (test)	0.155	The model explains 15.5% of score variation on unseen innings.
R^2 (train)	0.414	The model explains 41.4% of score variation on training data.

In plain terms: on average, our model's predictions were off by about **22 to 23 runs**. In a T20 match where teams score around 160 runs, this is a margin of roughly 14%. A broadcaster using this model to say "this team looks on track for 155" would be directionally correct, with the actual total typically falling somewhere between 132 and 178 runs.

The gap between training R^2 (0.414) and test R^2 (0.155) indicates the model has partially overfit to the training data. It learned some patterns specific to those 573 innings that do not fully generalise. This is expected behaviour and is addressed in the limitations section.

Prediction Scenarios

Two hypothetical Powerplay situations were passed through the trained model:

Feature	Strong Powerplay	Weak Powerplay
Powerplay runs	52	22
Powerplay wickets	1	3
Powerplay run rate	8.67	3.67
Powerplay sixes	3	0

Feature	Strong Powerplay	Weak Powerplay
Powerplay fours	6	1
Dot balls	12	20
Predicted Final Total	167 runs	132 runs

Both predictions feel realistic. A strong Powerplay of 52 runs with only one wicket lost predicts a final total of 167, which is above the dataset average of 160 and makes sense given the aggressive start. A struggling Powerplay of just 22 runs with three wickets lost predicts a total of only 132, which is well below average and consistent with what we observe in actual PSL matches.

5. Why Is the Model Limited?

The current model explains only **15.5%** of score variation on the test set, which means 84.5% of the variation in final totals is not captured by Powerplay statistics alone. There are two categories of limitation:

5.1 Data Volume

Although 717 innings is a reasonable starting point, it represents only the PSL competitions available in this dataset. Machine learning models for T20 score prediction typically require **500+ innings** to generalise well, and ideally more. With data from all PSL seasons sourced from cricsheet.org (500+ innings), the R^2 would typically reach 0.65–0.80, meaning the model would explain 65–80% of score variation. The pipeline built here is production-ready; it simply needs to be fed more data.

5.2 Missing Factors

The Powerplay statistics capture how a team started, but not the many other factors that shape the final total:

- Pitch and ground conditions: a dry, turning pitch in Karachi produces very different scores from a fast, flat pitch in Lahore. The model has no information about pitch behaviour.
- Weather and dew: evening matches with heavy dew make bowling slippery and difficult, often inflating second-innings totals beyond what the Powerplay suggests.
- Team form and confidence: a team on a winning streak bats with a very different mindset in the middle overs than a side under pressure. The model treats all innings identically regardless of tournament context.
- Individual player quality: the model does not know who is batting at numbers 4 through 7. A team with two explosive finishers can add 40–50 extra runs in the death overs regardless of the Powerplay score.
- Match situation in the second innings: a chasing team targeting a modest total of 130 may reach it with wickets in hand and stop, scoring far less than their batting potential would suggest.

For professional use, this model should be supplemented with live pitch reports, player availability data, and venue-specific historical performance.

6. Conclusion

This analysis confirms that the Powerplay is meaningfully linked to a team's final innings total in PSL T20 cricket. The Linear Regression model correctly identifies that higher Powerplay run rates, fewer wickets lost, and fewer dot balls are all associated with higher final scores, which makes strong cricket sense.

The model's average prediction error of 22.7 runs means it works well as a directional guide for identifying whether a Powerplay has set up a strong or weak innings, rather than as a precise forecasting tool. With a larger dataset covering more PSL seasons, prediction accuracy would improve substantially.

Key recommendation for a PSL coach: prioritise aggressive boundary-hitting in the first 6 overs while protecting top-order wickets. The data clearly shows that a strong Powerplay start is the single most controllable predictor of a competitive innings total.